thylinao at BAREC Shared Task 2026: Ensembling at the Noise Floor

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Abstract

Team thylinao entered the Open and Strict tracks of the BAREC 2026 shared task on sentence-level Arabic readability with a deliberately plain system: a fourteen-member equal-weight ensemble of encoder fine-tunes, decoded to the 19-level scale by cutpoints interpolated halfway between validation-tuned thresholds and thresholds that match the blind score distribution to the training label prior. It scored 85.3 QWK on both tracks, first on both. This paper is about how those choices were measured. Three independent estimates place the noise floor of our offline evaluation between 0.09 and 0.44 QWK, as large as the margins that decided the final ranking. In three paired comparisons against the live board, only the smallest of the three offline estimates survived; the other two reversed in sign. We therefore proved that the Open and Strict tracks score identical gold, and made every decision worth a submission from a live board reading. Once the board showed the mean was capped, every candidate we probed came back a tie, bar one that we later withdrew as outside the rules. The full ledger, including four refuted hypotheses, is in the appendix.

1 Introduction

Midway through the BAREC 2026 campaign, our ensemble scored 85.44, 85.28 and 85.42 on the same test split while we varied nothing but the resolution of a threshold search grid. Chasing that anomaly produced the central finding of this paper: measured three independent ways, the noise floor of our offline evaluation sits between 0.09 and 0.44 QWK, while the margins that decided our two placings were 0.1 and 0.3 QWK. An evaluation cannot choose between changes smaller than its own noise, and ours did not. In three paired tests, two of three offline estimates reversed in sign on the blind leaderboard. We responded by rebuilding the campaign around the one instrument with adequate resolution, first proving that the Open and Strict tracks score identical blind gold, then making every decision worth a submission from a live board reading. This paper describes the system that finished first on both tracks at 85.3 QWK, and the measurement discipline that chose it, which we believe is the part worth reusing.[1]

The first application of the rule came early. The public CAMeL-Lab readability checkpoints were fine-tuned on the BAREC corpus, which contains the shared task's training split, so their training-split predictions are in-sample: solo QWK of 89 to 95 on train against 82 to 84 on clean holdouts. Any blend weight or cutpoint tuned on train would have been tuned on memorised labels, so we retired train as a tuning surface, tuned on validation, and kept test as a holdout we read rather than fitted.

We contribute a noise-floor audit of offline evaluation on this task, with three independent estimators and three paired offline-versus-board tests on the true blind gold; a protocol for using a second track as a measurement instrument, including a cross-track parity proof and two competition mechanics that changed our decisions more than any model change did; a cutpoint calibration that was the one lever clearing the noise floor, worth 84.1 to 84.5 on the board; and an audited ledger of four refuted hypotheses and the improvements the ensemble absorbed without moving (Appendices A and B).

2 Background

BAREC grades Arabic sentences on a 19-level ordinal scale aligned to a pedagogical reading progression (Elmadani et al., 2025b; Habash et al., 2025). The shared task evaluates systems with Quadratic Weighted Kappa (Elmadani et al.,

[1] Code, the fine-tuning grid and the per-run score arrays: https://github.com/thylinao1/barec-2026-readability

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2026). QWK penalises an error by the square of its distance in levels, so a system is judged on the shape of its prediction distribution at least as much as on per-sentence accuracy. That drove most of what we did, and it is visible in the final standings: two teams finished with far higher exact accuracy than ours and far lower QWK.

We entered the sentence-level Open and Strict tracks. Three results from the 2025 edition shaped our plan. Ensembling across loss families was the winning margin: the winner (Basem et al., 2025) combined four encoders under three loss families and reported 84.8 for its best single model rising to 87.5 for the ensemble, and the runner-up stacked seven regression base learners under a meta-classifier (Saeed et al., 2025). Ordinal framing and D3Tok morphological tokenisation are table stakes, worth roughly +3.1 and +1.0 on the organisers' own baselines, and other entries explored ordinal heads, uncertainty-aware decoding and QWK surrogates (Nazzal, 2025; Abdou, 2025; Trigui, 2025; Mohamed Emad Elden, 2025). External data mostly did not help, and the organisers say so directly (Elmadani et al., 2025a): the winning system scored worse in Open than in Strict. That is why our Open and Strict systems are identical. We treated the Open track not as a licence to add outside corpora but as a second measurement of the same blind gold, which turned out to be worth more. Earlier Arabic readability resources include the OSMAN metric, a formula in the Flesch and Fog tradition with Arabic syllable counting (El-Haj and Rayson, 2016), the SAMER 26,000-lemma levelled lexicon and simplification corpus for school-aged learners (Al Khalil et al., 2020; Alhafni et al., 2024), and DARES, 13,335 instances from Saudi school textbooks (El-Haj et al., 2024); BAREC differs in grading single sentences on 19 levels.

3 System Overview

3.1 Members and blend

The final system averages fourteen z-standardised member scores at equal weight, standardising each member's validation and blind scores independently before averaging. Four seats are public BAREC fine-tunes released by others, nine are seed-averaged families of our own fine-tunes and one is a single run. Ours use the BAREC training split alone (Section 5.5). Members came from a cluster grid of 112 completed fine-tuning runs over 13 backbones, 6 loss families and three input views, combined with public checkpoints into a pool of 172 candidates (Appendix D).

3.2 The measurement chain

Before training anything we reimplemented the official scorer and gated the project on exact agreement, reproducing the official script string-for-string on the development fixture. A separate entry point pins the label set to $[1, 19]$ for per-fold evaluation, where a sparse tail would otherwise silently build a smaller confusion matrix. The gate paid for itself in one upload: a submission whose local test QWK was 86.10 returned 86.10 on the development-phase board, which scores that released test split, all seven metrics matching, confirming that our grader is the board's grader.

Preprocessing parity is the same kind of concern. The blind set is new sentences, so the corpus's gold morphological views must be recomputed locally with CAMeL Tools (Obeid et al., 2020). Two failure modes here are silent: feeding raw text to a model that expects the Word view drops that member from 84 QWK to 50, and omitting only the character-normalisation step inside that pipeline drops exact match against gold from 99% to 31%. Neither raises an error; both produce a plausible submission.

The subtler version cost us a member. Our locally computed D3Tok matches the released gold at about 76%, because the released column used a licence-gated morphological database. Measured naively this appeared to drop the regression member from 84.68 to 82.75, so we shipped a word-only ensemble. That was wrong, and the error was in the standardisation, not the pipeline: the comparison took the member's validation scores from the gold column while taking its blind scores from ours, and z-scoring cannot reconcile two input distributions. Recomputing validation through the same pipeline as blind made the member self-consistent, and adding it back was worth +0.3 on the board, more than every later calibration experiment combined. A member's standardisation statistics and the scores they are applied to must come from one pipeline, end to end.

3.3 Half-match cutpoint calibration

Cutpoints tuned on validation put the blind predictions in the wrong place. The system predicted level 12 for about 8% of blind sentences where the corpus prior puts roughly 21%. The scores shift

between validation and blind, so cutpoints fitted in validation score space land wrongly in blind score space, and quadratic weighting punishes the resulting systematic level shift.

The correction uses the only label-free target available, the training label histogram. Let r be the validation-tuned cutpoint vector, p the training label prior over levels 1 to 19, and s_b the blind blended scores. We ship

$$c = \mathrm{sort}\big((1-f)\,r + f\,Q(s_b, \mathrm{cumsum}(p)_{1:18})\big) \tag{1}$$

with $f = 0.50$, where Q takes empirical quantiles of s_b at the cumulative prior boundaries. The shipped cutpoints sit halfway between what maximised QWK on validation and what would make the blind prediction histogram match the corpus prior.

We measured f on the live board, because blind has no labels and the test split does not exhibit the shift that motivates the correction. The curve is single-peaked: $f = 0$ gives 84.1, $f = 0.50$ gives 84.5, $f = 0.60$ gives 84.4, and $f = 1$ gives 84.0. Both endpoints are worse than the midpoint, by 0.4 and 0.5, because the two targets carry complementary and partly wrong information: the rounder knows the score-to-level mapping but assumes the wrong score distribution, while the prior match knows the right output distribution but discards the rounder's ordering information. After five readings we stopped tuning f.

4 Experimental Setup

The release provides train (54,845 sentences over 1,518 documents), validation (7,310) and test (7,286) with gold labels, and a blind test of 8,077 sentences over 212 documents. Folds are built once with grouped stratified 5-fold cross-validation, grouped on document so topic and vocabulary do not leak, and serialised to a committed fold map that is never regenerated. Our go/no-go number is a nested out-of-fold QWK in which cutpoints are fitted on the other folds and applied to the held-out fold, so thresholds never see the rows they score. Every member shows validation below test by about 1.3 QWK, which correctly predicted the blind set would score nearer 85 than 86. Appendix F states what each split was used for.

Calibrating the instrument. We estimated the noise of that protocol three ways. Holding data, members and weights fixed and varying only the threshold search grid between 150 and 500 candidates moves the same blend over a range of 0.345 QWK with standard deviation 0.094. Freezing the cutpoints and bootstrap-resampling the 7,286 test rows 400 times puts the sampling standard deviation of a single test measurement at about 0.44 QWK. Varying the fold-map seed, by contrast, returns bit-identical results, because the rounder snaps cutpoints to a fixed grid and small changes in the fitting rows often do not move the selected grid point. That third result is a trap worth naming: anyone estimating variance by reshuffling folds on a grid-search rounder will conclude, incorrectly, that their procedure is deterministic. Vary the grid instead. A paired bootstrap shrinks a naive +0.52 gain, from adding MARBERTv2 to the three-member word-only blend under tuned weights, to +0.24 with a 95% interval of $[+0.03, +0.45]$.

The second track as an instrument. Two byte-identical files submitted to Open and to Strict returned identical metric vectors, so both tracks score the same 8,077-row blind gold. This makes the Open board a free measurement instrument for a Strict system whenever the Open runner-up is far enough behind that a losing probe costs no place. Two mechanics constrain the protocol, and both cost us before we understood them. First, we believed Codabench displays each participant's latest submission; the competition API showed a submission rule of `Add`, so the board shows the entry the participant explicitly added and probing is close to free. Second, the board orders equal QWK by exact accuracy, so only a strictly better score should be sent to a track we lead.

5 Results

5.1 Offline estimates against the board

Table 1 pairs every offline estimate we acted on with the live board reading of the same change. Two of the three reversed in sign, and the only change that delivered was the one with the smallest offline estimate.

The precision of an offline protocol is not its accuracy. Our nested out-of-fold measurements carry a standard deviation of 0.019 across grid settings, which invites reading the +0.406 MARBERTv2 gain as twenty standard deviations of signal; the board returned -0.1. That 0.019 is the reproducibility of the procedure, not the uncertainty of the quantity it estimates. Nested cross-

change	offline	board
add MARBERTv2 (nested OOF)	$+0.406$	-0.1
document shrinkage α=0.1	$+0.22$	-0.2
add D3Tok regression member	$+0.086$	$+0.3$

Table 1: Paired offline-versus-board comparisons. Each row is one change, estimated offline and then measured on the blind gold.

validation controls threshold overfitting and says nothing about distribution shift between the tuning and target samples, which is what separated validation from blind.

5.2 Saturation

Once the ensemble reached fourteen members it absorbed everything. Against an equal-weight baseline of 85.281 on honest nested out-of-fold, twenty fitted-aggregation configurations scored between 84.74 and 85.12: non-negative and ordinary least squares weights with ridge-shrunk and clipped variants, direct-QWK coordinate ascent, an expanded 44-member pool, per-member isotonic calibration, and median, trimmed-mean and rank-mean aggregation (Appendix C). Every one is below equal weight. Validation cannot support fourteen fitted parameters, and the row-wise extremes that robust aggregators discard carry real ordinal signal.

Member-level gains were absorbed too: a word-to-D3Tok lexicon worth +0.91 QWK solo moved the blend by between -0.024 and $+0.001$, because the cleaned member correlates more with the pack (Appendix C).

5.3 The endgame

With the mean provably capped, we changed the selection criterion. A dual gate, requiring a candidate to beat the baseline both on honest nested out-of-fold and on a deployment-mirrored oracle over the released gold test split, called four staged candidates statistical ties; probed on the board they returned 85.2, 85.2, 85.2 and 85.3, four for four at the board's resolution. So we stopped ranking candidates by offline score and ranked them by disagreement with the incumbent. Five near-clones, differing on 2% to 12% of rows, drew ties. The one candidate that changed 10.0% of predictions drew 85.4, the only reading above the incumbent in the campaign, and it is the entry we withdrew (Section 5.5). The test leg of the gate never overturned validation: its one promoted candidate failed the validation leg, and the board agreed.

5.4 Final standing

The official results, published after the correction of Section 5.5, rank us first on both tracks with 85.3 QWK and 37.7 exact accuracy (85.27 and 37.75 in the organisers' rescoring, Appendix A). Open lists 4 teams and its runner-up scored 85.2; Strict lists 8 and its runner-up 85.0. The runner-up's exact accuracy exceeded ours on both tracks, 39.6 against 37.7 in Open, which our pool could not reach by recombination, calibration or retraining.

5.5 Correction to the submitted system

The entry we first submitted had four of its fourteen seats fine-tuned on the training split plus the corpus's released gold test split. We disclosed this to the organisers and in this paper but had read the rule wrongly: it says models are trained exclusively on the training set. The organisers said so after the phase closed and we withdrew the entry. Every number for our system is the same ensemble with those four seats replaced by their training-split-only counterparts from the same grid, an entry also submitted during the testing phase, which the board scored at 85.3. What it bought is measured in Appendix C: 0.008 offline and 0.1 on the board, inside every noise estimate in Section 4.

6 Conclusion

Three conclusions, in ascending order of how uncomfortable they are. First, a nested out-of-fold estimate is not automatically an honest estimate; ours was correctly specified and still overpromised by 0.5 QWK. Second, the reported standard deviation of a repeated measurement is not the uncertainty of what it measures. Third, on this task the leaderboard was the only instrument with adequate resolution, and the second track let us use it without risk. That last point needs two tracks scoring the same gold and a margin wide enough to make a losing probe free, so teams without one should budget submissions as measurements rather than as attempts.

If a seven-thousand-row holdout cannot resolve 0.2 QWK, some fraction of the ablation gains published for QWK-scored tasks, possibly including some in this paper, are reports of noise. Teams with board access should publish paired offline-versus-board comparisons.

Limitations

Our noise-floor evidence rests on three paired comparisons, enough to show that the disagreement occurs but not how often. Board readings are single measurements on a fixed 8,077-row set and carry their own sampling noise, which we could not estimate because we cannot resample the blind gold. The second-track oracle depends on a margin over the runner-up and was unavailable to us once that margin closed. Our locally computed D3Tok is systematically degraded relative to a licensed team's, and the standardisation fix recovers the member's usefulness without recovering that loss. Finally, equal weights and a single calibration scalar are unrefuted at our measurement resolution rather than shown optimal; we cannot rule out that a better combiner exists below our noise floor.

Two of our practices depart from a single-shot test protocol, and a reader who holds to that protocol should discount them: cutpoints are adjusted to the blind prediction histogram, and among offline ties the shipped candidate was chosen by leaderboard readings. Both are label-free or mediated by the board within the task's submission limit, and Appendix F lists every such reading so that the departure is auditable, but it is a departure, and the released test split served as a second holdout that we read repeatedly rather than as one final measurement.

Ethical Considerations

BAREC frames sentence-level readability along a pedagogical reading progression (§2), so a system like ours is most plausibly consumed downstream to grade or select reading material for learners. We have only evaluated our ensemble on the shared task's own train, validation, test and blind splits; we do not know how it performs on registers, dialectal admixture, or genres outside that distribution, and a deployment decision (including or excluding a text for a given reader) should not extend our reported numbers to a population the corpus does not represent. The BAREC corpus and annotation guidelines we build on are cited in §2 and used under their released research licences; the D3Tok licensing asymmetry discussed above under Limitations is a data-access fairness concern between shared-task participants, not a claim about harm to end users of a deployed system.

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A Submission ledger

Every row is a live board reading on the blind gold. Rows marked with a dagger were probes uploaded to the Open track and read but never added to a leaderboard, which the competition's `Add` submission rule makes free (Section 4).

The organisers' rescoring of the shipped file, sent to us on 4 September 2026 after the phase

system	calibration	QWK
word-only 3-member	$f = 0$	84.1
half-match champion	$f = 0.50$	**84.5**
match-60	$f = 0.60$	84.4
full distribution match	$f = 1.00$	84.0
metadata-reweighted prior	$f = 0.50$	83.9
blend4reg (both tracks)	$f = 0.50$	**84.8**
blend5, adds MARBERTv2†	$f = 0.50$	84.7
blend4reg + doc shrinkage†	$f = 0.50$	84.6
fleet11	$f = 0.50$	85.1
fleet10strict	$f = 0.50$	85.2
fleet14, shipped, both tracks	$f = 0.50$	**85.3**
fleet14 tie probes, 4 files†	$f = 0.50$	85.2 to 85.3
fleet14tt4, withdrawn	$f = 0.50$	85.4

Table 2: Blind-phase submissions, in the order they were measured.

closed and after their ruling of 2 September that the first entry fell outside the rules, is QWK 85.27, exact accuracy 37.75, adjacent accuracy 70.82, average absolute distance 1.136, and 60.51 / 66.93 / 74.59 at the 7, 5 and 3 collapsed levels. It rounds to the board's 85.3 and 37.7. The official results on the task website, announced to participants on 5 September 2026, list this line first on both tracks.

Two rows in our internal log recorded a track that we could not re-confirm after the phase closed, and one calibration reading at $f = 0.40$ came from an ambiguous screenshot. The authenticated submission history is not reachable through the public API, so rather than guess we dropped the $f = 0.40$ point and describe the calibration peak as flat between 0.5 and 0.6 rather than across 0.4 to 0.6.

B Negative results in full

Metadata-reweighted label prior, refuted at 83.9. The half-match calibration matches the blind scores to the raw training label prior. That target is arguably wrong, because the blind set's composition differs sharply from train: the STEM domain rises from 6.9% to 17.8% and the Specialized text class from 26.9% to 44.7%, while Advanced falls from 33.0% to 19.3%. Domain and text class are present for 100% of blind rows and take values that exist in train, which permits a label-free reweighting $P(\ell) = \sum_s P(s \mid \text{blind})\, P(\ell \mid s, \text{train})$ over $s = \text{Domain} \times \text{Text_Class}$. This shifts the prior by L_1 distance 0.1224 and alters 1,697 of 8,077 predictions, every one by exactly $+1$ level. We built it as a strictly single-variable change: the implementation's own raw-prior reference output is byte-identical to the previously submitted champion, so the prior is provably the only thing that differs. It scored 83.9, losing 0.6.

The hypothesis could not be tested offline, because blind has no labels and the test split does not exhibit the shift that motivated the correction. What the refutation rules out is specific: pushing 21% of predictions up one level makes things worse, so the blind set is not systematically harder in readability level than train despite being 2.6 times more STEM. We suspect this is because BAREC levels are annotated against a reading progression rather than topic difficulty, so domain composition and level composition are less coupled than intuition suggests, but we did not test that explanation.

MARBERTv2 as a member, measured at -0.1. MARBERTv2 was the most attractive remaining candidate on paper: a different backbone family, and among the least correlated with our blend at 0.945 against 0.95 to 0.98 for everything else. Offline it measured $+0.41$ on the clean test holdout and $+0.406$ on honest nested out-of-fold, the largest gain in either table. The test-holdout figure did not even reproduce offline: rerun from the cached arrays the baseline came out at 85.28 and the blend with the member at 85.28, a gain of exactly zero. Recovering it was expensive, because the original fine-tuning script saved only prediction arrays and never called `save_pretrained`, so the model existed only as numbers and had to be retrained. On the board it cost -0.1 in the five-member configuration, the only configuration containing it that we submitted.

Document-level score shrinkage, measured at -0.2. The blind set ships a document column with 212 documents over 8,077 sentences, which invites smoothing each sentence score toward its document mean, $s \leftarrow (1-\alpha)s + \alpha\, \text{mean}_{\text{doc}}(s)$. On the clean test holdout the effect is single-peaked and sharp: 85.48 at $\alpha = 0$, 85.69 at 0.1, 85.32 at 0.2, 84.88 at 0.3, and a collapse to 82.15 at 0.5. We shipped the peak and it cost 0.2. One implementation subtlety has to be right for the transform to be the one that was measured: shrinkage must be applied to both validation and blind scores, because the rounder's cutpoints live in validation score space.

Weak or redundant members. AraELECTRA regression is the most decorrelated candidate we measured at 0.9101, but at 78.29 solo it is too weak

to help and moves the blend from 85.07 to 84.87. A D3Tok cross-entropy member costs -0.055 on honest nested out-of-fold and was excluded even though its sibling regression member is our best single addition. Decorrelation alone does not qualify a member; our working rule is that a member must be either near 85 solo or below 0.90 correlated, and one that is neither is not worth a seat.

C The saturation sweep

All figures are honest validation nested out-of-fold QWK, mean over threshold grids of 200, 300 and 400 candidates. The equal-weight fourteen-member baseline is 85.281 ± 0.051.

aggregation	QWK
equal weight (shipped)	**85.281**
NNLS weights, ridge-shrunk 25/50/75	84.79 to 85.10
OLS and clipped weights	84.79 to 85.10
direct-QWK coordinate ascent	84.97, 84.98
expanded 44-member pool, NNLS	84.94 to 85.10
per-member isotonic calibration	84.74 to 85.12
median / trim1 / trim2 / rank-mean	84.79 to 85.01

Table 3: Twenty aggregation configurations, all below plain equal weight. Per-fold fitted weights were unstable, zeroing members in some folds and not others.

Member-level improvements were absorbed as well. We built a word-to-D3Tok lexicon from the gold corpus alignment (98.8% of sentences align), covering 92% of validation tokens and 91% of blind tokens, with the older disambiguation pipeline filling the remainder. This raised sentence-exact fidelity against the gold column from 63.7% to 79.1% and lifted the affected regression member from 82.27 to 83.18 solo on validation, recovering most of the gap to gold text. At blend level the same change was worth -0.019 when swapped for the single member, $+0.001$ for the seed-averaged family, and -0.024 for both together. We also retrained the five strongest families on the training split plus the released gold test split, 62,131 rows and a 13% increase in gold data. The blend we submitted took four of the five and moved from 85.281 to 85.289; taking all five instead gives 85.091. That entry was withdrawn as outside the rules (Section 5.5) and is reported here because it is the cleanest evidence of saturation we have. On the board the withdrawn file scored 85.4 against the shipped 85.3, on predictions differing for 811 of 8,077 sentences. The gap is as small on all seven official metrics, listed in the order QWK, exact and adjacent accuracy, average absolute distance and the 7, 5 and 3 level accuracies: the withdrawn file's board line was 85.4 / 37.8 / 71.3 / 1.1 / 60.1 / 67.1 / 74.8, and the organisers' rescoring of the shipped file, rounded to the same precision, reads 85.3 / 37.7 / 70.8 / 1.1 / 60.5 / 66.9 / 74.6. No metric moved by more than half a point at the board's precision, and the seven-level accuracy is the only one that rose.

D Fine-tuning grid

We used no data outside BAREC in either track, and every model we fine-tuned saw only the BAREC training split. We used no external readability resources: the SAMER corpus (Alhafni et al., 2024) and the SAMER readability lexicon (Al Khalil et al., 2020), listed for the Constrained and Open tracks, were not used, and we added nothing outside BAREC in the Open track either, since the 2025 evidence is that external data does not help on this task.

Training used a Slurm job array on A100 GPUs, with the single exception of the MARBERTv2 regression member discussed in Appendix B, which was fine-tuned in a free Kaggle notebook on a T4 and is not part of the cluster grid. One trainer script takes backbone, loss, input view, seed, learning rate and warmup ratio, and writes validation and test score arrays through the same code path used at inference. Defaults: learning rate 2×10^{-5}, 5 epochs, batch 16, maximum sequence length 128, no warmup except 0.06 for the large backbones. Cross-entropy, soft-label and focal heads are decoded to a continuous score by the posterior mean $\mathrm{softmax}(z) \cdot [1..19]$; regression heads use the raw output; CORN follows Shi et al. (2023).

The dispatched grid has 112 rows over 13 backbones and 6 loss families (cross-entropy 45, regression 44, soft-QWK 10, focal 6, CORN 5, soft-label 2), crossing three input views (raw, Word, D3Tok) with seeds. All 112 completed with a done marker and both score arrays; a smoke-test run brings the total to 113. With public checkpoints and seed-averaged families this produced a pool of 172 candidate members, 26 of them seed averages. The thirteen backbones are AraBERT (Antoun et al., 2020), AraELECTRA (Antoun et al., 2021), CAMeLBERT (Inoue et al., 2021), QARiB (Abdelali et al., 2021), ARBERTv2 and MARBERTv2 (Abdul-Mageed et al., 2021), XLM-R (Conneau et al., 2020) and mDeBERTaV3

origin	heads / inputs	seats
public checkpoints	CE, reg; Word, D3Tok, raw	4
own AraBERT v02/v2/large	CE, reg; Word, D3Tok	4
own CAMeLBERT, QARiB	CE, reg; Word, raw	3
own MARBERT, AraELECTRA, XLM-R	CE, reg; raw, Word	3

Table 4: The fourteen seats, by origin. Nine are seed-averaged families of our own fine-tunes, four are public checkpoints and one is a single run. Every seat we fine-tuned uses the BAREC training split; the four public checkpoints were fine-tuned on BAREC by others. The exact member list is given below.

(He et al., 2023).

Two gates apply before a member may enter a blend, both added after an incident in which a checkpoint export was omitted and only prediction arrays survived. The head shape must match the fine-tune rather than the base model, and locally recomputed validation scores must reproduce the cluster's saved arrays above 0.99 correlation. A member failing either gate is not blended, because a standardisation computed from mismatched scores is silently wrong rather than loudly wrong.

The fourteen seats of the final system are: the four public-checkpoint champion members (`arabertv02-word-CE`, `camelbert-word-CE`, `qarib-barec-optimized-v8`, `arabertv2-d3tok-reg` scored through our own D3Tok pipeline); seed-averaged families of `arabertv2-reg-d3tok`, `arabertv02-ce-word`, `alarge02-reg-word` and `marbert-ce-raw`; seed-averaged families of `camelbert-ce-word`, `qarib-ce-raw`, `araelectra-reg-word`, `camelbert-reg-word` and `arabertv2-ce-d3tok`; and a single `xlm-roberta-large` regression run. The four public checkpoints are `CAMeL-Lab/readability-arabertv02-word-CE`, `CAMeL-Lab/readability-camelbert-word-CE` and `CAMeL-Lab/readability-arabertv2-d3tok-reg`, released with the BAREC corpus (Elmadani et al., 2025b), and the community checkpoint `AymanTarig/qarib-barec-optimized-v8`, which has no associated publication and documents no training data.[2] That checkpoint entered our candidate pool through a screening script that scored each public candidate on validation and on the released gold test split, discarded anything above 0.88 on either as memorised, and retained only those above 0.80 on test. Its presence in the pool was therefore conditioned on a test reading. Nothing downstream is fitted on test, and a validation-only greedy over the compliant pool selects it first of 62 candidates, at 85.275 against 85.253 for the runner-up, so the gate is not what put it in the system.

Every seat we fine-tuned ourselves uses the BAREC training split alone. The four that did not, in the entry we first submitted, were replaced by their training-split-only counterparts from the same grid after the organisers ruled on them (Section 5.5). The nine seed-averaged families and the single run each carry a per-run training configuration on disk, and none names any split other than train. The four public checkpoints were fine-tuned on BAREC by others: three are the corpus authors' own baselines, whose paper describes fitting on train and selecting on dev, and the fourth is a community checkpoint whose author documents no training data at all, which we report because we cannot verify it. Cutpoints and member selection are fitted on the validation split, the corpus's development set, which the organisers confirmed is within the rules, since the corpus has no separate tuning split.

E Harness detail

The rounder maps a continuous score to a level in $[1, 19]$ via 18 monotone cutpoints chosen by coordinate ascent to maximise QWK. The search is seeded from equal-frequency quantiles of the label distribution, each cutpoint is swept over a fixed grid spanning the score range, monotonicity is enforced between neighbours, and the output is clipped. We use monotone coordinate ascent rather than Nelder-Mead, which can return crossed, non-monotone thresholds in 18 dimensions and is retained only as a cross-check. On a classical ridge baseline the protocol reads cleanly: naive rounding 71.25, nested out-of-fold 74.48, non-nested out-of-fold 74.63.

Folds are built once with grouped stratified 5-

[2] https://huggingface.co/AymanTarig/qarib-barec-optimized-v8

fold cross-validation, grouped on document and stratified on the 19-level label, and serialised to a committed document-to-fold map covering 1,518 documents. The map is loaded identically on every machine and never regenerated, because regenerating it per machine is how out-of-fold arrays silently fail to line up. All five folds carry all 19 levels.

The parity gate compares our scorer against the official development fixture at the level of the printed string: QWK 80.0040%, accuracy 56.6211%, adjacent accuracy 69.8632%, average absolute distance 1.143776, and 65.8687% / 70.2736% / 76.4569% at 7, 5 and 3 collapsed levels.

F Data use by split

Train. Fine-tunes every member we trained, and supplies the label prior p in Equation 1. It is not a tuning surface: the public checkpoints were fine-tuned on the BAREC corpus, which contains this split, so their train predictions are in-sample, at solo QWK of 89 to 95 against 82 to 84 on clean holdouts, and any weight or cutpoint tuned on train would be tuned on memorised labels.

Validation. Fits the 18 cutpoints, selects members by forward selection run nested out-of-fold, and yields the go/no-go estimate of Section 4 and every figure in Appendix C.

Released test split, with gold. Fitted nothing in the shipped system. It was read in six ways. The development-phase board scored it, so during that phase a local test QWK equalled the board reading, which is how the parity gate of Section 3 was confirmed with one upload. It is reported beside validation for every member. The bootstrap noise estimate of Section 4 resamples its rows. It was the test leg of the dual gate in Section 5, whose one promoted candidate failed the validation leg. A screening script admitted public checkpoints to the candidate pool on it, which conditioned one shipped seat's presence in the pool (Appendix D). Finally, two refuted changes in Appendix B, MARBERTv2 and document shrinkage, were estimated on it before being measured on the board, and the shrinkage strength $\alpha = 0.1$ was chosen from its curve.

Blind set, no labels. Supplied the score histogram that Equation 1 matches to the training prior, and the Domain and Text_Class composition behind the refuted reweighted prior in Appendix B. Every other use is a leaderboard reading, within the task's limit of five submissions per day: the calibration readings that fixed f (Section 3), the three paired comparisons of Table 1, and the tie probes of Section 5, with the ledger of submissions in Appendix A. Among offline ties, the shipped system was chosen by those readings.

G Reproducibility

Code, the fine-tuning grid, and the per-run validation and test score arrays are at `https://github.com/thylinao1/barec-2026-readability` under an MIT license. Rebuilding the final submission from the committed blender and the member score arrays reproduces the submitted file byte for byte; its SHA-256 is `cf2c752406a9ea2bc0ad94e83f40190cab87b958512ef9c03dcf01c12768b762`. The member arrays computed over the blind input are withheld together with everything else derived from the confidential blind set, so the rebuild command in the repository README needs either those arrays or the blind input to regenerate them.

Predictions over the blind test set are deliberately not published. The blind set is released only to registered participants and may be reused in a future edition, so a strong prediction file over it would function as a pseudo-label source for later participants. The code regenerates those files for anyone holding the blind input, and the hash above lets the organisers verify the submitted file exactly.